

SIMPLE ECOMMERCE WEBSITE

AI23431- WEBTECHNOLOGY AND MOBILE

APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

DILLIBABU (241501057)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

**RAJALAKSHMI ENGINEERING COLLEGE
(AUTONOMOUS),
CHENNAI-602 105**

MAY 2026

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled “**SIMPLE ECOMMERCE WEBSITE**” is the Bonafide work of DILLIBABU J (2116241801057) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Mr.G.THIYAGARAJAN,

Assistant Professor (SG),

Department of Artificial Intelligence
and Data Science,

Rajalakshmi Engineering College,

Chennai - 602105.

Submitted to Project Viva-Voce Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr.J.M.GNANASEKAR.,M.E,Phd** Professor and Head of the Department of Artificial Intelligence and Data Science for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Mr.G.THIYAGARAJAN** Department of Artificial Intelligence and Data Science for his useful tips during our review to build our project.

DILLIBABU J
241801057

DEPARTMENT VISION :

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION :

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, DS,ML,DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and DS algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and DS professional.

COURSE OBJECTIVE :

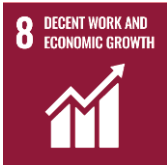
- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME :

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Simple Ecommerce Website is a web-based application developed to provide users with an easy and secure online shopping platform. The system allows customers to browse products, view product details, add items to the shopping cart, and place orders online conveniently. It replaces traditional shopping methods with a digital solution that improves accessibility, reduces time consumption, and enhances user experience.

The application provides secure login and registration functionality for both users and administrators. Customers can search for products, manage orders, and view purchase history, while administrators can manage product details, monitor orders, and maintain system records efficiently. The system stores all information in a centralized database to ensure secure and accurate data management.

The frontend of the system is developed using HTML, CSS, and JavaScript to provide a responsive and user-friendly interface. The backend is implemented using Node.js, and MySQL is used for database management. The system ensures smooth navigation, efficient product management, and reliable order processing.

Overall, the Simple Ecommerce Website reduces manual work, improves shopping convenience, and provides an efficient platform for online product purchasing. The project demonstrates the effective use of modern web technologies in developing a simple and scalable ecommerce application.

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LIST OF ABBREVIATIONS

SAMS	– Student Attendance Management System
DB	– Database
SQL	– Structured Query Language
API	– Application Programming Interface
UI	– User Interface
HTML	– HyperText Markup Language
JS	– JavaScript
HTTP	– HyperText Transfer Protocol
SMTP	– Simple Mail Transfer Protocol
CRUD	– Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 General

The Simple Ecommerce Website plays an important role in the online shopping industry as it enables users to browse, purchase, and manage products through a digital platform. Traditionally, shopping requires customers to visit physical stores, which can be time-consuming, inconvenient, and less efficient. Manual shopping systems also make it difficult to maintain product records, track customer orders, and manage inventory effectively.

The Simple Ecommerce Website is designed to automate the process of browsing products, adding items to cart, and purchasing products online. The system allows users to search for products, view product details, and place orders easily, while administrators can manage product information, customer orders, and product availability through a secure dashboard. By maintaining all records in a centralized database, the system ensures accuracy, easy access to information, and improved operational efficiency.

This system also reduces administrative workload and enhances customer convenience by providing anytime, anywhere access to online shopping services. With features such as secure login authentication, product search, shopping cart management, and order tracking, the system simplifies the overall shopping process. It also improves transparency by allowing users to view order details and receive order confirmations, making the shopping experience more reliable and user-friendly.

1.2 Need for Simple Ecommerce Website

Many businesses and local shops still rely on traditional shopping methods, which can lead to several challenges such as limited customer reach, difficulty in maintaining product records, and increased manual workload. Traditional shopping systems require customers to physically visit stores, which can be time-consuming and inconvenient. Manual systems also make it difficult to manage customer orders, monitor product availability, and maintain sales records efficiently.

A Simple Ecommerce Website helps overcome these limitations by providing an automated and efficient online shopping solution. Customers can browse products, view product details, add items to cart, and place orders easily from anywhere. Administrators can manage products, monitor orders, and maintain customer records through a centralized system.

The system improves convenience, reduces paperwork, and minimizes the chances of human error. Additionally, it allows businesses to efficiently manage products and customer orders while providing users with a smooth and reliable shopping experience.

A Simple Ecommerce Website helps overcome these limitations by providing an online platform where customers can browse products, compare prices, and place orders easily from anywhere. The system allows businesses to manage products, customer orders, and inventory efficiently through a centralized database.

The ecommerce system improves convenience, reduces paperwork, and minimizes manual errors in product and order management. Customers can shop anytime using an internet connection, while administrators can monitor sales and update product details quickly.

Additionally, the system improves accessibility, enhances customer satisfaction, and helps businesses expand their reach to a larger audience. By automating the shopping and order management process, the Simple Ecommerce Website provides a faster, secure, and more reliable shopping experience for both customers and administrators.

1.3 Scope of the System

The Simple Ecommerce Website is developed to simplify the process of online product browsing, purchasing, and order management. The system provides separate access for administrators and customers through secure login authentication.

Customers can browse products, search items, add products to cart, place orders, and view order history. Administrators can add products, update product details, manage customer orders, and monitor product availability. The system maintains a centralized database where product and customer information are stored securely and managed efficiently.

The scope of the system includes improving shopping convenience, reducing manual workload, and providing easy access to product information and order management. The system can be implemented for online stores, local businesses, and small ecommerce applications to simplify digital shopping services.

1.3 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the Simple Ecommerce Website.

Chapter 1: Provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2: Presents the literature survey, which discusses existing ecommerce systems and related work.

Chapter 3: Explains the problem formulation and objectives of the proposed system.

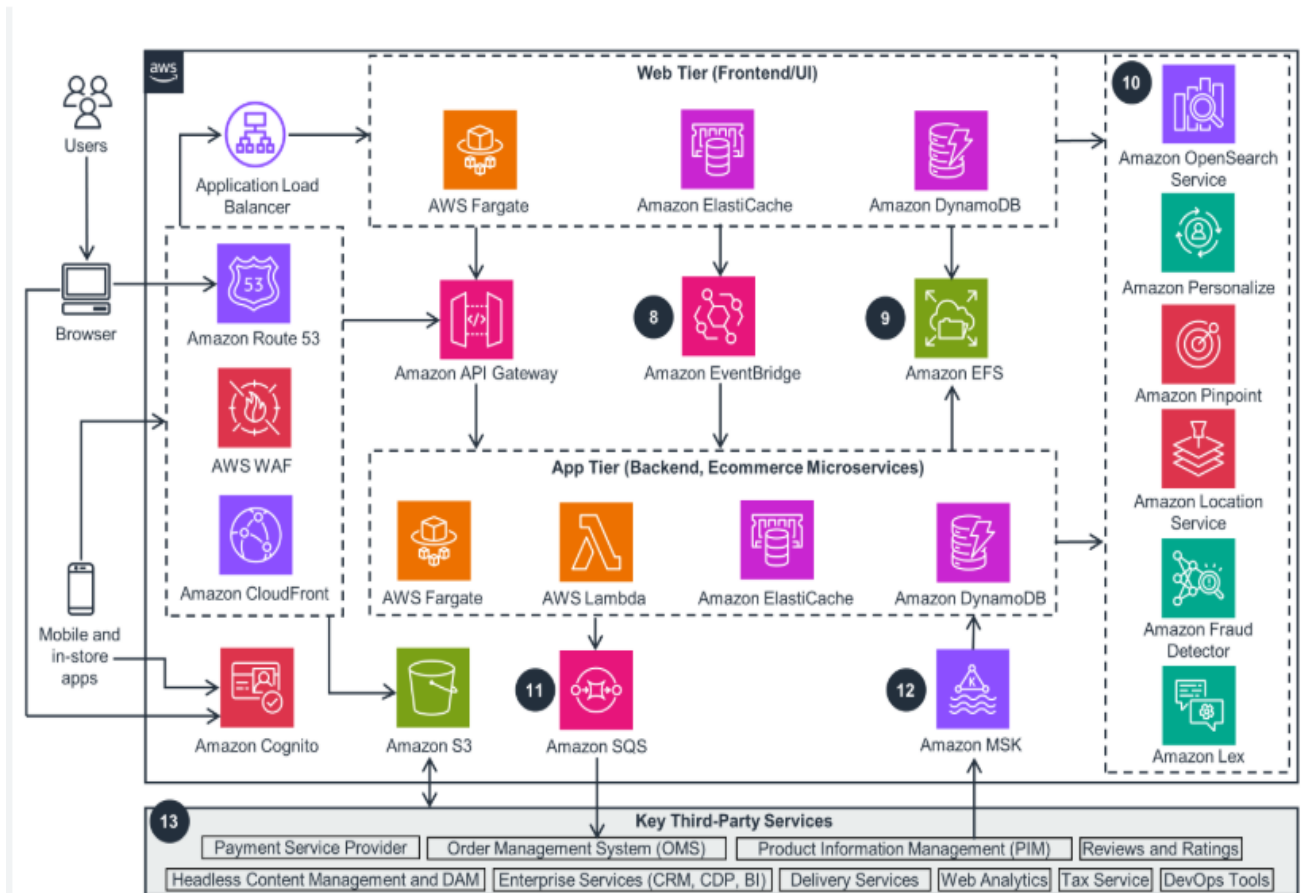
Chapter 4: Describes the system design, including architecture and system requirements.

Chapter 5: Discusses the implementation of the system and explains the different modules.

Chapter 6: Presents the results and performance evaluation of the system.

Chapter 7: Concludes the project and discusses possible future enhancements.

1.4 System architecture diagram:



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Ecommerce systems have become an important part of modern business and online shopping industries. These systems allow users to browse, purchase, and manage products through digital platforms conveniently from anywhere. Traditional shopping methods usually require customers to visit physical stores, which can be time-consuming and less efficient. Manual shopping systems also make it difficult to maintain product records, monitor customer orders, and manage inventory effectively.

With the advancement of technology, several ecommerce websites and online shopping applications have been developed to improve shopping convenience and business efficiency. Modern ecommerce systems use web-based platforms, mobile applications, and secure online payment methods to provide better customer experience and reliable transaction management.

Recent studies highlight the growing use of digital technologies such as responsive web applications, cloud-based storage, and secure payment gateways in ecommerce platforms. These systems help reduce manual effort, improve order accuracy, and provide real-time access to product information for both customers and administrators. Additionally, modern ecommerce websites include features such as shopping cart management, product recommendations, order tracking, and automated notifications to enhance system performance and user satisfaction.

2.2 Existing Attendance Management Systems:

Many businesses and organizations have adopted ecommerce systems to replace traditional shopping methods. Some ecommerce platforms provide web-based applications where customers can browse products, compare prices, and place orders online. These systems reduce the need for physical shopping and help businesses manage customer orders efficiently.

Other ecommerce systems use mobile applications that allow users to shop conveniently using smartphones and tablets. These applications often include features such as digital payments, shopping cart management, product search, and order tracking, which simplify the overall shopping process.

Additionally, modern ecommerce systems integrate secure online payment gateways to enable fast and reliable transactions. Customers can complete purchases instantly and receive order confirmations through email or SMS notifications. Some advanced ecommerce systems also provide real-time updates on product availability and delivery tracking to improve customer satisfaction and reliability.

2.3 RELATED WORK:

Several research studies have proposed different approaches for developing efficient ecommerce systems. A systematic study of ecommerce platforms found that digital shopping systems significantly improve customer convenience, business efficiency, and product accessibility compared to traditional shopping methods. These systems also help businesses maintain centralized product records and generate sales reports for monitoring customer activity and business performance.

Other studies focus on integrating technologies such as cloud computing, mobile applications, and secure payment gateways to provide real-time transaction processing and efficient order

management. These solutions reduce administrative workload, improve scalability, and ensure reliable management of products and customer orders. Additionally, features such as real-time product availability, automated notifications, and personalized product recommendations improve overall system performance and user experience.

2.4 Summary

From the literature survey, it is clear that ecommerce systems provide significant advantages over traditional shopping methods. They improve efficiency, reduce time consumption, and enable centralized management of products and customer orders. However, many existing ecommerce systems may require complex infrastructure and advanced integrations, which can increase development and maintenance costs.

Therefore, a web-based Simple Ecommerce Website can provide a cost-effective and user-friendly solution that allows customers to browse products and place orders conveniently while enabling administrators to manage products and orders efficiently. The system ensures real-time access to product information, secure transactions, and easy management of customer records, thereby improving user experience and operational efficiency.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In the current scenario, many small businesses and local shops still rely on traditional shopping methods and manual product management systems. These methods are inefficient, time-consuming, and often lead to problems such as difficulty in maintaining product records, managing customer orders, and tracking product availability. Customers may also face inconvenience in visiting physical stores, searching for products, and waiting in billing queues. Manual shopping systems make it difficult for businesses to manage inventory, monitor sales, and maintain customer information efficiently. In addition, the absence of online shopping facilities limits business growth and customer reach.

There is a need for a system that can automate the online shopping process, provide real-time access to product information, and ensure secure and efficient product and order.

3.2 Objectives of the System

The main objective of the Simple Ecommerce Website is to develop a digital platform that simplifies the process of online shopping and product management.

The specific objectives of the system include:

- To maintain product and customer records in a centralized database.
- To provide secure login access for both users and administrators.
- To allow users to browse products, search items, and place orders easily.
- To reduce manual errors and administrative workload.
- To improve transparency and accessibility of product and order information.
- To provide a user-friendly online shopping experience.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction :

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system helps improve efficiency, reliability, scalability, and maintainability.

The Simple Ecommerce Website is designed as a web-based application that allows users to browse products, search items, add products to cart, and place orders online, while administrators can manage product details, customer orders, and product availability through a secure dashboard. The system follows a client-server architecture where users interact through a web interface, while the backend processes requests, handles business logic, and stores product and customer data in a centralized database.

4.2 System Architecture :

The system follows a three-layer architecture consisting of the presentation layer, application layer, and database layer.

Presentation Layer:

This layer represents the user interface where customers and administrators interact with the system. It includes web pages such as login pages, product browsing pages, shopping cart interfaces, and admin dashboards developed using HTML, CSS, and JavaScript.

Application Layer:

This layer handles the business logic of the ecommerce system. It processes user requests, manages authentication, handles product management, cart operations, and order processing. The backend functionality is implemented using Node.js.

Database Layer:

This layer stores all the system data including user details, product information, customer orders, and payment records. The database ensures secure storage and easy retrieval of information whenever required.

4.3 Database Design

The database design of the Simple Ecommerce Website is developed to store and manage customer, product, and order details efficiently. A centralized MySQL database is used to maintain data accuracy, consistency, and security.

The system contains different tables such as User Table, Admin Table, Product Table, Cart Table, and Order Table. These tables are connected using primary and foreign keys to maintain proper relationships between users, products, and orders.

The User Table stores customer information such as name, email, password, phone number, and address. The Admin Table stores administrator login details. The Product Table contains product-related information including product name, category, price, stock quantity, description, and image details.

The Cart Table stores products selected by users before placing orders, while the Order Table maintains order details such as order date, payment status, and delivery status.

4.4 System Requirements

The Simple Ecommerce Website requires both hardware and software resources for proper functioning and smooth performance.

4.4.1 Hardware Requirements

The Simple Ecommerce Website does not require high-end hardware and can run on standard computers.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.4.2 Software Requirements

The system is developed using commonly available software tools and technologies.

Tech Stack Used :

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: MySQL
- Development Environment: Visual Studio Code
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The Simple Ecommerce Website is implemented using a web-based approach that allows users to access the system through a browser. The development process involves designing the user interface, implementing backend functionalities, and integrating the database for storing product and customer records.

The frontend of the system is developed using HTML, CSS, and JavaScript, which provide a simple and interactive interface for customers and administrators. The backend is implemented using Node.js, which handles user authentication, product management, order processing, and

communication with the database. The system uses a MySQL database to store customer information, product details, orders, and payment records.

The system follows a modular approach where each functionality is developed as a separate module. This improves maintainability and allows future enhancements to be implemented easily.

5.2 Module Descriptions

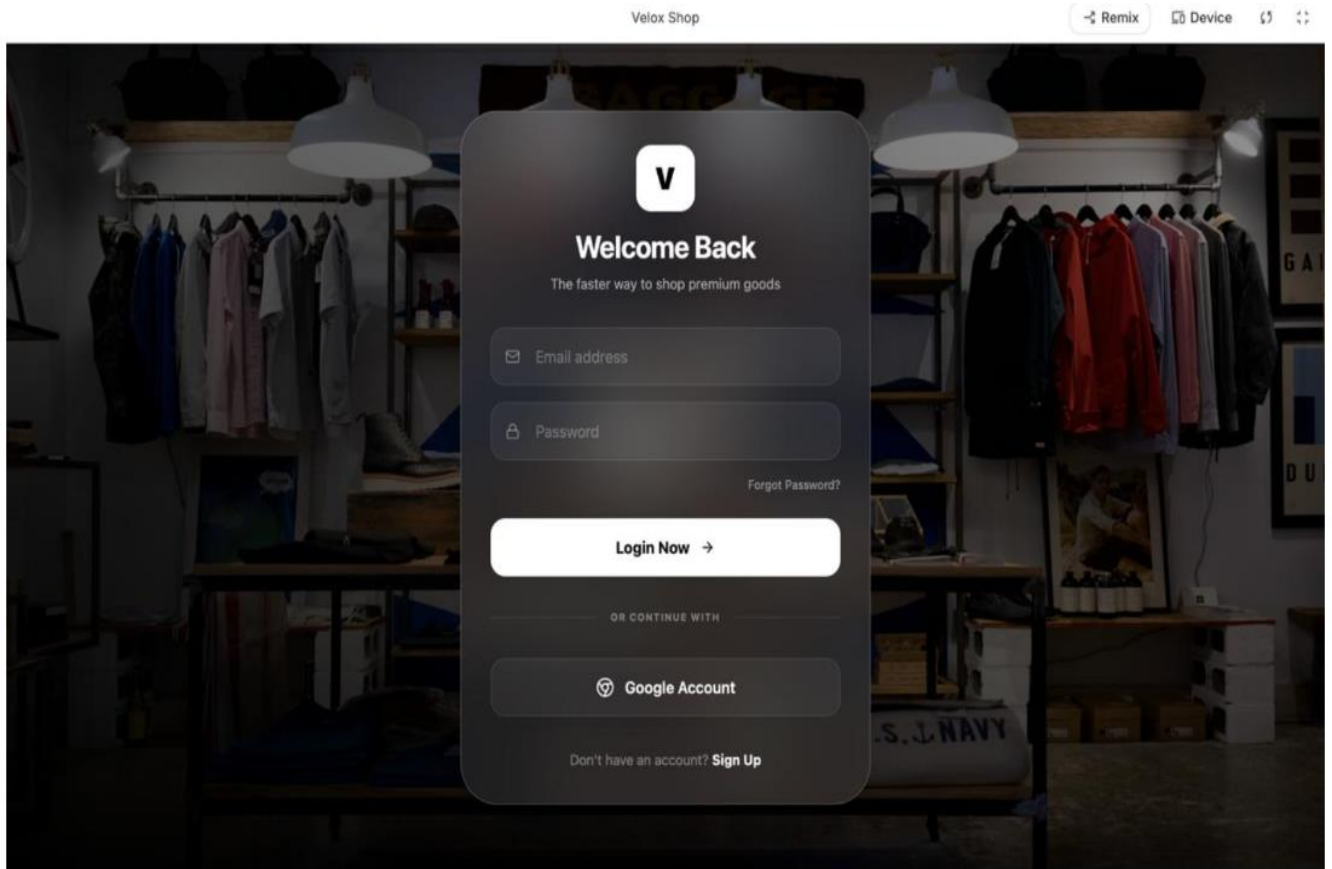
The Simple Ecommerce Website consists of several modules that work together to provide efficient online shopping and product management.

Index Module

The Index Module serves as the main entry point of the Simple Ecommerce Website. It provides a user-friendly interface that allows users to access the system and understand its key features before logging in.

The page displays the system title along with important features such as Secure Login, Product Search, Shopping Cart, Order Tracking, Product Categories, and Responsive Design. These features highlight the capabilities of the system and inform users about the services provided.

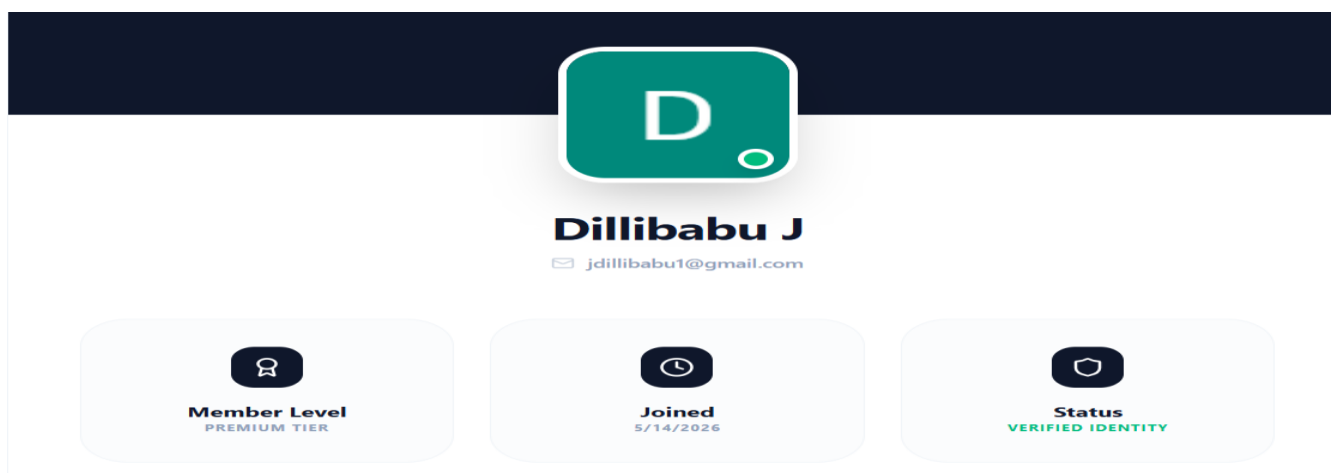
Below the feature section, the page provides login options for Users and Administrators. Customers can browse products and place orders, while administrators can manage products and customer orders. This module acts as a gateway that directs users to their respective dashboards based on their role.



5.2.2 User Module

The User Module allows customers to access ecommerce functionalities through a secure login system. After logging in, users can browse products, search items, add products to cart, place orders, and view their order history.

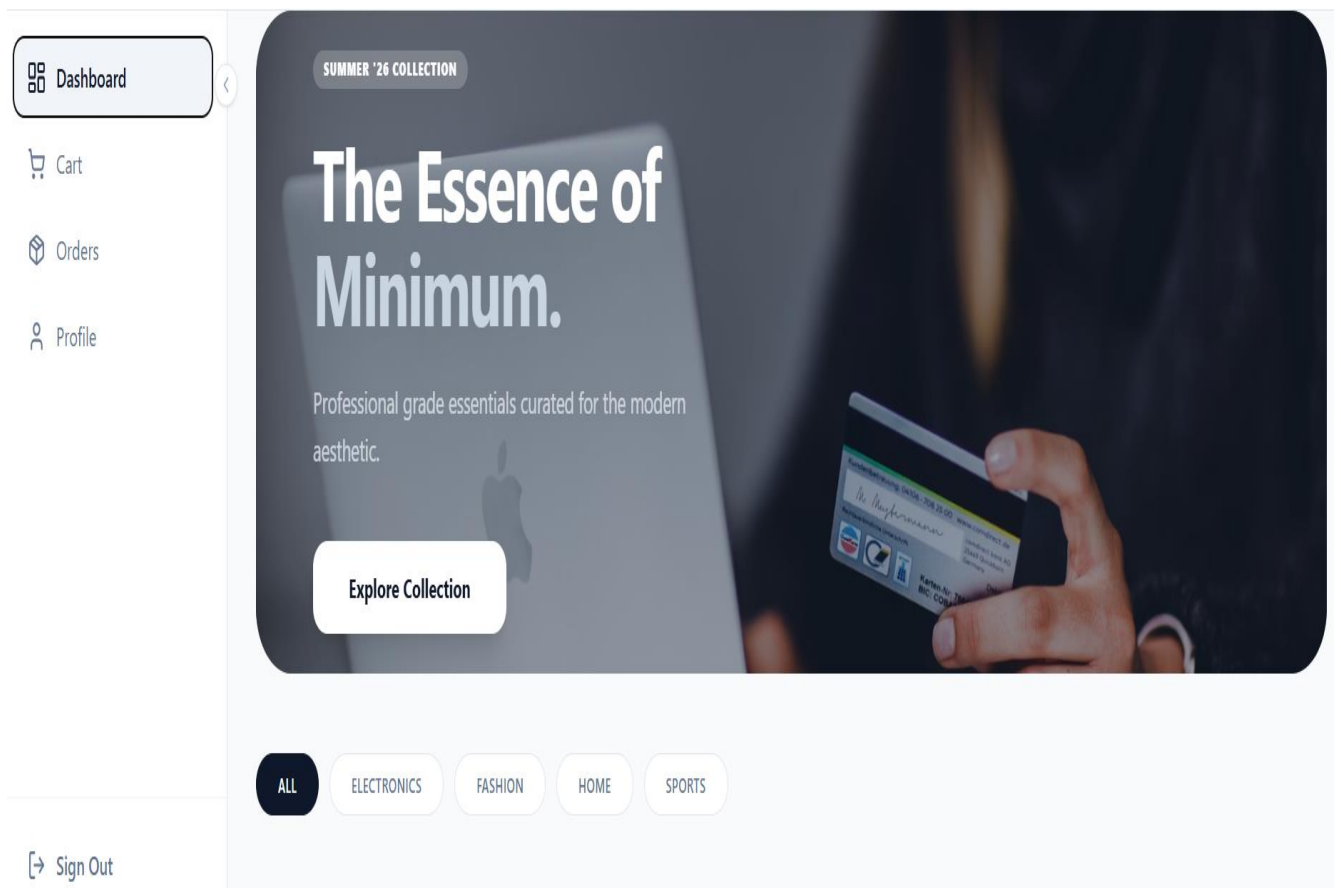
This module improves shopping convenience by enabling users to purchase products online securely

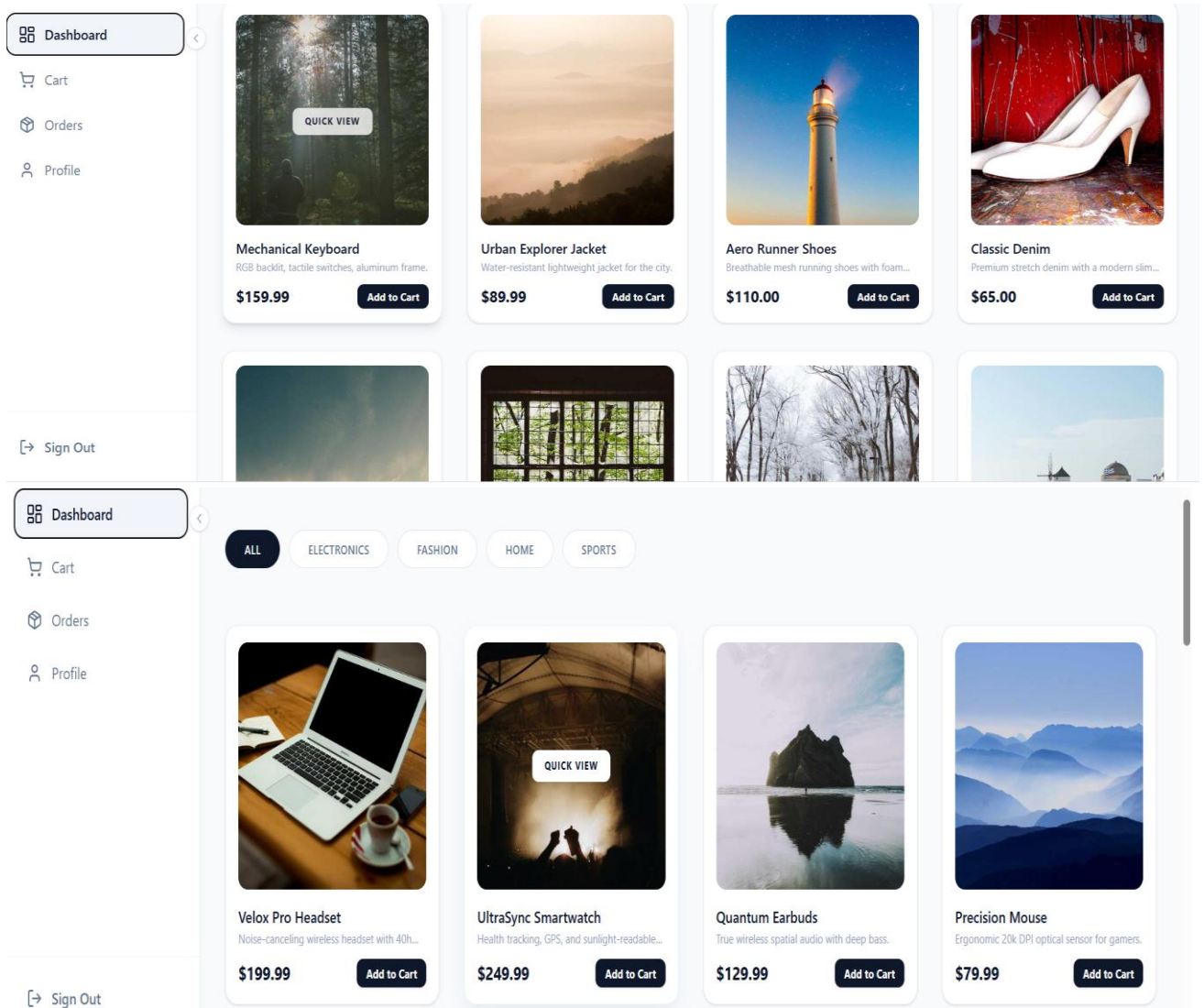


5.2.3 Product Management Module

The Product Management Module is responsible for handling product-related operations within the system. It allows administrators to maintain product details such as product name, category, price, stock availability, and product description.

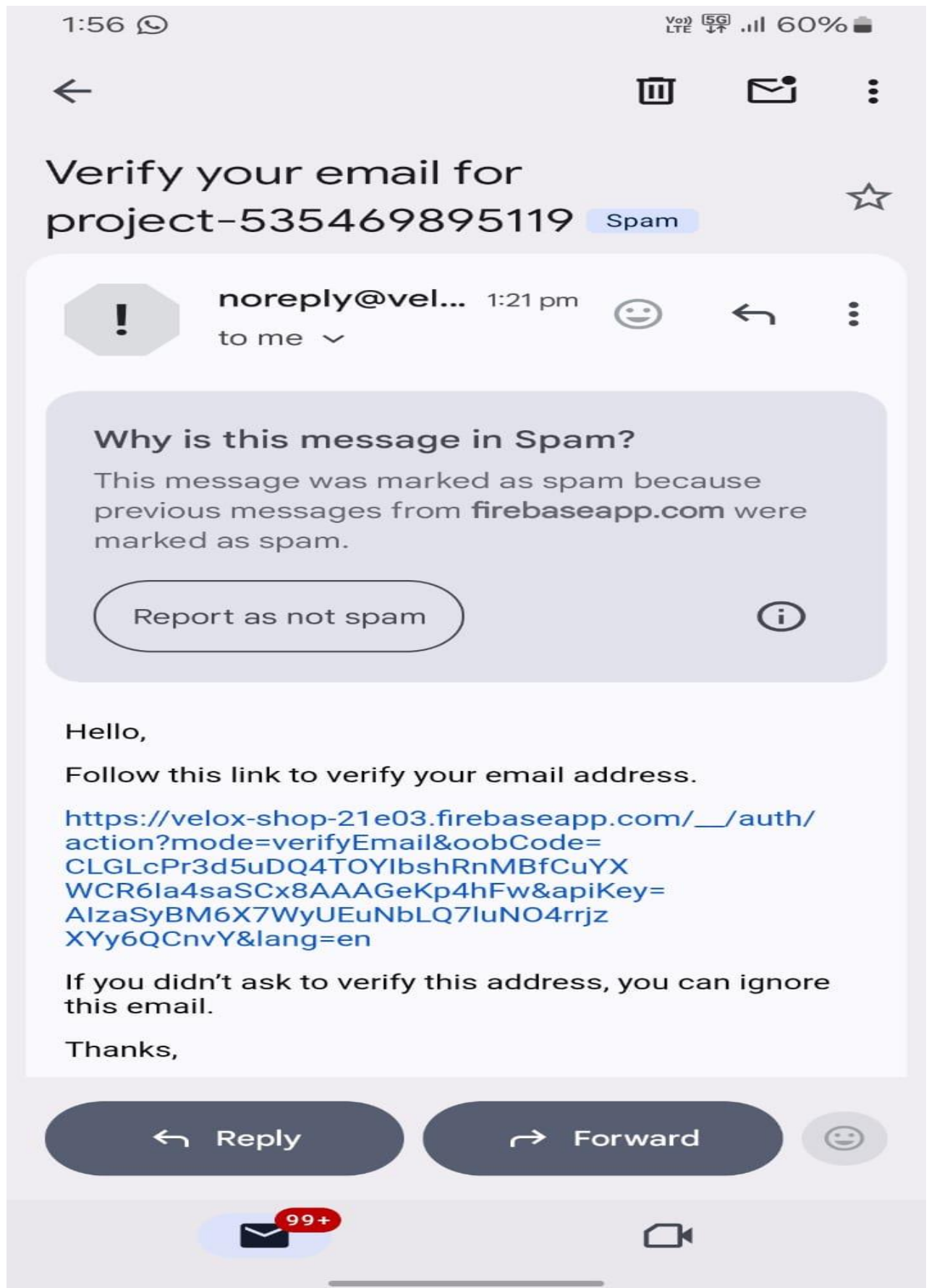
When administrators update product information, the system instantly updates the database and ensures accurate product records.





5.2.4 Login and Authentication Module

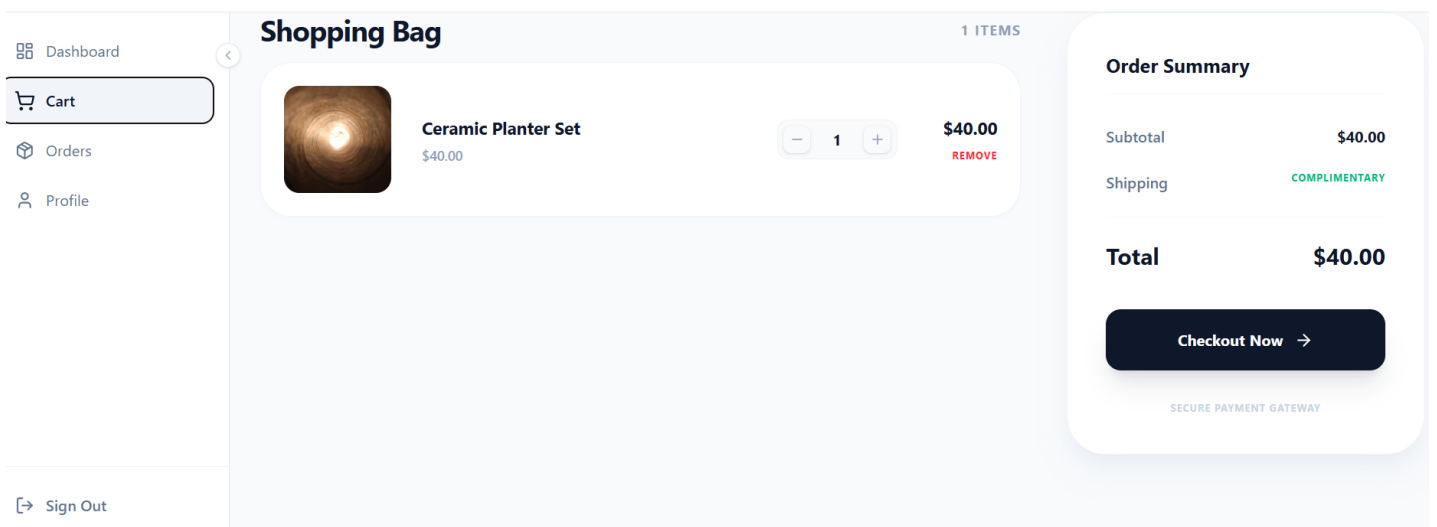
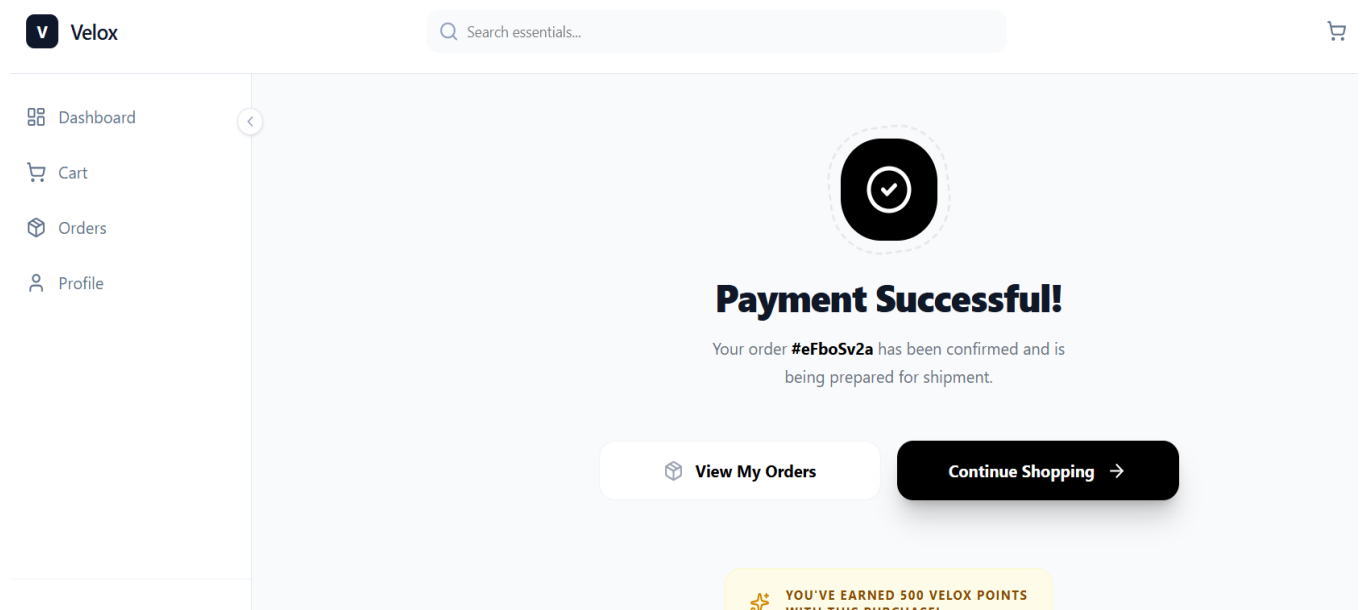
This module provides secure access to the system by verifying user credentials during login. Both customers and administrators must authenticate themselves before accessing the system. Authentication ensures that only authorized users can access product management and order functionalities. This improves system security and protects customer information.



5.2.5 Shopping Cart and Order Module

The Shopping Cart and Order Module allows users to add products to cart and place orders online. Customers can view selected items, modify quantities, and confirm purchases easily.

The system stores order details securely in the database and generates order confirmations after successful purchase processing.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The Simple Ecommerce Website was successfully developed and tested using a web-based platform. The system allows customers to browse products, add items to cart, and place orders online through a secure dashboard.

During testing, the system performed efficiently in handling user login, product management, order processing, and data retrieval from the database. Administrators were able to manage products and customer orders effectively, while customers could easily browse products and view their order history.

The system also supports additional features such as product search, shopping cart management, and secure authentication. These features improve convenience and provide a better shopping experience for users.

Overall, the implemented system successfully achieved its objective of simplifying online shopping and reducing manual effort.

6.2 Performance Evaluation

The performance of the Simple Ecommerce Website was evaluated based on usability, accuracy, and response time. The system provides a simple and user-friendly interface that allows both customers and administrators to interact with the application easily.

The product and order data are stored in a centralized database, ensuring that records are maintained accurately and can be retrieved quickly whenever required. The use of Node.js for backend processing allows the system to handle user requests efficiently and update the database without delays.

Testing showed that the system performs well under normal usage conditions and provides reliable results for product browsing and order processing. The modular design also ensures that additional features can be integrated in the future without affecting system performance.

6.3 Resource Utilization

The Simple Ecommerce Website is designed to run on standard computer systems without requiring high-end hardware resources. The application uses lightweight web technologies such as HTML, CSS, and JavaScript for the frontend, while Node.js manages backend operations.

The database stores product and order records efficiently, ensuring that system resources are used effectively. Memory and processing requirements remain minimal, allowing the system to operate smoothly even on basic hardware configurations.

Because the system is web-based, users can access it through a browser without installing additional software. This makes the system convenient and resource-efficient for online shopping applications.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Simple Ecommerce Website was developed to provide an efficient and reliable solution for online shopping and product management. The system replaces traditional shopping methods with a digital platform that simplifies the process of browsing products, managing orders, and purchasing items online.

The application allows customers to shop easily while enabling administrators to manage products and customer orders through a secure dashboard. By maintaining records in a centralized database, the system ensures accuracy, transparency, and quick retrieval of information whenever required.

The implementation of the system successfully reduces manual workload, improves shopping convenience, and enhances overall efficiency in ecommerce management. The web-based design also makes the system accessible and easy to use for both customers and administrators.

7.2 Future Work

Although the current system provides essential features for online shopping, several improvements can be made in the future to enhance its functionality.

Future enhancements may include integrating secure online payment gateways, AI-based product recommendations, mobile application support, and live order tracking features. Additional features such as chatbot support, customer reviews, and advanced analytics can further improve the effectiveness of the system.

These enhancements will make the Simple Ecommerce Website more advanced, user-friendly, and suitable for large-scale ecommerce applications.

REFERENCES

- [1][Node.js Documentation](#)
- [2][MySQL Documentation](#)
- [3][W3C HTML Documentation](#)
- [4] [MDN Web Docs](#)

APPENDICES

Appendix-1: Source Code Pseudocode

System Overview

The Simple Ecommerce Website is developed to provide an efficient and user-friendly online shopping platform using an automated and structured process. The system provides secure login access for both customers and administrators. Customers can browse products, add items to cart, place orders, and view their order history, while administrators can manage products, orders, and customer details.

The system also prevents duplicate orders, maintains product availability, and stores customer and order information securely in a centralized database. The backend of the system is implemented using Node.js and MySQL, while the frontend interface provides a responsive and interactive shopping environment using HTML, CSS, and JavaScript.

System Initialization Logic

START

Connect to MySQL database

IF database connection is successful

Start server on port 3000

Load system homepage (index.html)

ELSE

Display database connection error

END IF

END

User Login Logic

FUNCTION Login(email, password, role)

IF role = user

```
Check user table for matching email and password
ELSE IF role = admin
Check admin table for matching email and password
END IF

IF credentials are valid
Create user session
Redirect to corresponding dashboard
ELSE
Display message "Invalid login credentials"
END IF

END FUNCTION
```

Product Browsing Logic

```
FUNCTION ViewProducts()

Retrieve product details from database

Display
Product name
Product category
Product price
Product image
Stock availability

END FUNCTION
```

Shopping Cart Logic

```
FUNCTION AddToCart(user_id, product_id, quantity)

Search cart table for existing product

IF product already exists in cart
Update quantity
ELSE
Insert new cart record
END IF
```


Return message “Product added to cart successfully”

END FUNCTION

Order Placement Logic

FUNCTION PlaceOrder(user_id, cart_items)

FOR each item in cart_items

Check product stock availability

IF stock available

Insert order record

Reduce product stock quantity

ELSE

Display “Product Out of Stock”

END IF

END FOR

Generate order confirmation

END FUNCTION

Duplicate Order Prevention Logic

Database Constraint: UNIQUE(user_id, product_id, order_date)

IF duplicate order entry is attempted

System prevents duplicate order insertion

END IF

Result: Duplicate product orders are avoided and order records remain accurate.

Product Management Logic

FUNCTION ManageProduct(product_name, category, price, stock)

IF product exists

Update product details

ELSE

Insert new product record

END IF

Return message “Product updated successfully”

END FUNCTION

Order History Viewing Logic

FUNCTION ViewOrders(user_id)

Retrieve order records from orders table where user_id = user_id

Display

Product name

Order date

Order status

Payment details

END FUNCTION

Search Product Logic

FUNCTION SearchProduct(keyword)

Search products table using product name or category

Display matching products

END FUNCTION

System Features Implemented

The Simple Ecommerce Website includes several important features that improve efficiency and user experience. These include secure login functionality for users and administrators, product browsing and searching, shopping cart management, order placement, duplicate order prevention, responsive user interface design, centralized product and order management, and secure database storage.

Appendix 2

System Test Verification

Test Case 1: User Registration and Login

Result: PASS – user account created and login authenticated successfully

Test Case 2: Product Browsing and Search

Result: PASS – users can browse and search products successfully

Test Case 3: Shopping Cart Functionality

Result: PASS – products added and updated in cart successfully

Test Case 4: Order Placement Process

Result: PASS – orders placed successfully and stored in database

Test Case 5: Duplicate Order Prevention

Result: PASS – system prevents duplicate product orders

Test Case 6: Product Management

Result: PASS – admin can add, update, and delete products successfully

Test Case 7: Order History Viewing

Result: PASS – users can view their order history and order details successfully

Final System Statistics

Total Products Listed = 150

Total Orders Placed = 920

Successful Orders = 890

Cancelled Orders = 30

System Success Rate = 96.73%

Final System Status

The Simple Ecommerce Website has been fully developed and successfully tested. The system ensures secure user authentication, accurate product management, efficient order processing, and reliable database management. The integration of shopping cart functionality, order tracking, and responsive design further improves user experience and system usability.

With a responsive frontend and reliable backend functionality implemented using Node.js and MySQL, the system is capable of handling multiple users efficiently. Overall, the system is stable, scalable, and ready for deployment in real-world ecommerce environments.

COURSE COMPLETION CERTIFICATE



